

Coastal Turbidity Change in Waters Adjacent to Indonesia Morowali Industrial Park

Project Overview

Located on the coast of Central Sulawesi, the Indonesia Morowali Industrial Park underwent rapid expansion during the study period following the start of nickel pig iron production in 2015. Industrial development and land disturbance across Morowali's erosion-prone lateritic landscape have raised concerns about sediment runoff into nearby coastal ecosystems. This project evaluates changes in the optical turbidity signal of adjacent coastal waters by comparing Sentinel-2 Multispectral Instrument (MSI) imagery acquired on 2 April 2016 and 6 April 2021.

An automated ArcPy workflow applies Sentinel-2 Scene Classification Layer (SCL) masking, delineates water using the Normalized Difference Water Index (NDWI), and calculates the Normalized Difference Turbidity Index (NDTI) from red and green spectral values within the identified water pixels. Water masks from the two observation dates are then intersected to isolate persistent water for the multi-date NDTI change analysis. The analysis identifies widespread positive NDTI change, with the strongest increases concentrated along sections of shoreline adjacent to the industrial park.